

Push-button Switch & Buzzer Circuit

Let's Build!



What Will We Learn Today?

1. What is a push-button switch?
2. What is a buzzer?
3. How to connect them in a circuit
4. Build it yourself!

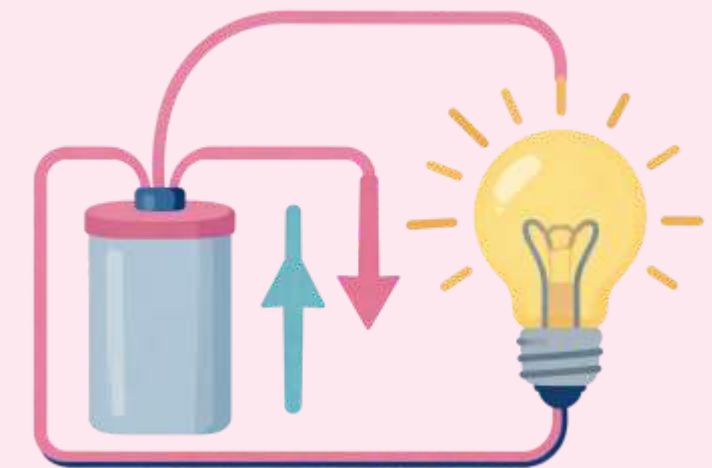


What is a Circuit?

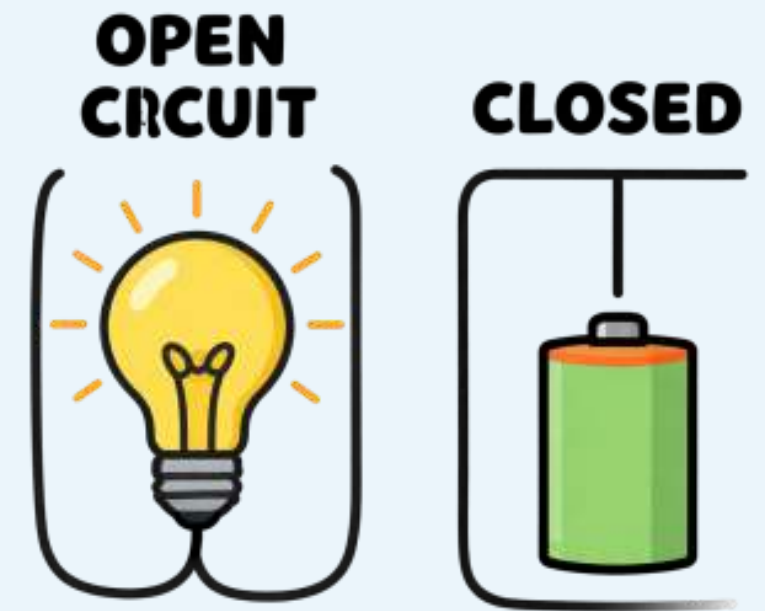
A circuit is a path that electricity flows through.

It needs: a power source, wires, and a component.

When the path is complete, electricity flows!



Open vs Closed Circuit



Open circuit = gap in the path = NO flow

Closed circuit = complete path = electricity flows!

A switch controls open and closed.

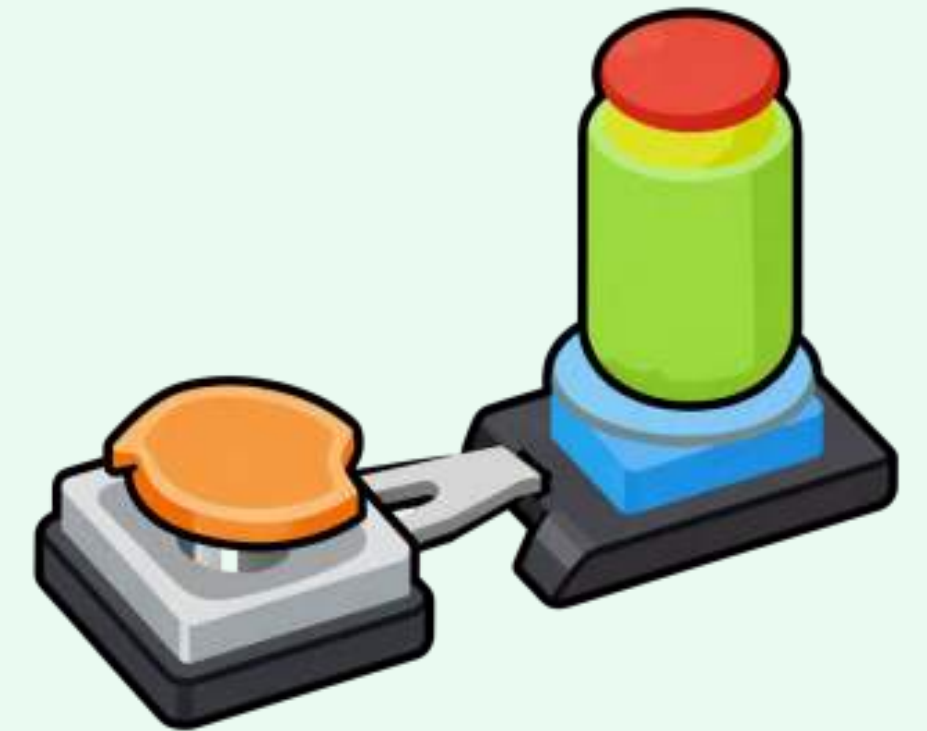
Meet the Push-button Switch!

A push-button switch closes the circuit
only when you PRESS it.
Let go = circuit opens again!



How Does a Push-button Work?

Inside are two metal contacts.
Press: contacts touch = closed circuit.
Release: contacts separate = open circuit.



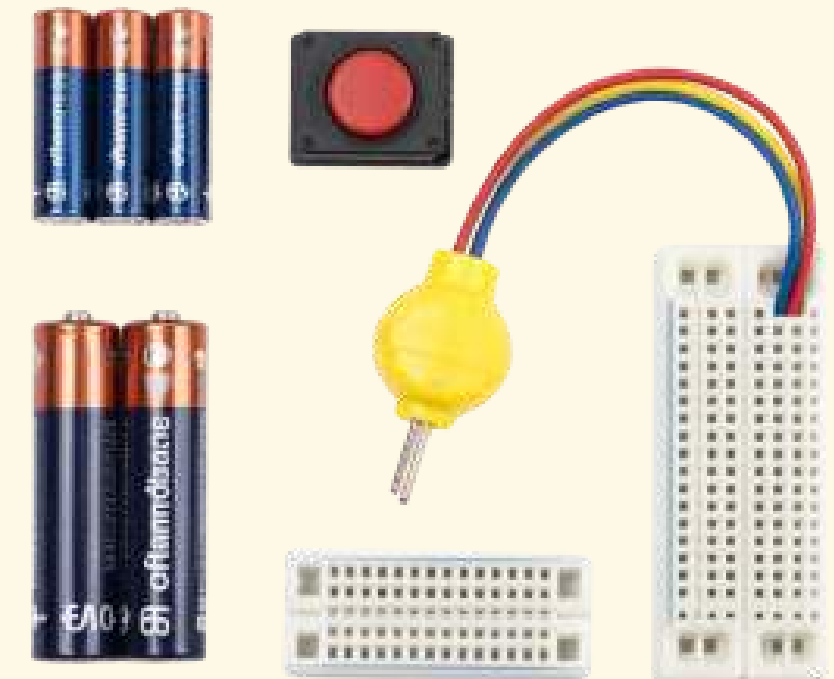
Meet the Buzzer!

A buzzer makes a sound when electricity flows through it.
It converts electrical energy into sound energy!



Parts We Need

1. Battery (3V or 9V)
2. Push-button switch
3. Buzzer
4. Connecting wires
5. Breadboard (optional)



Circuit Diagram

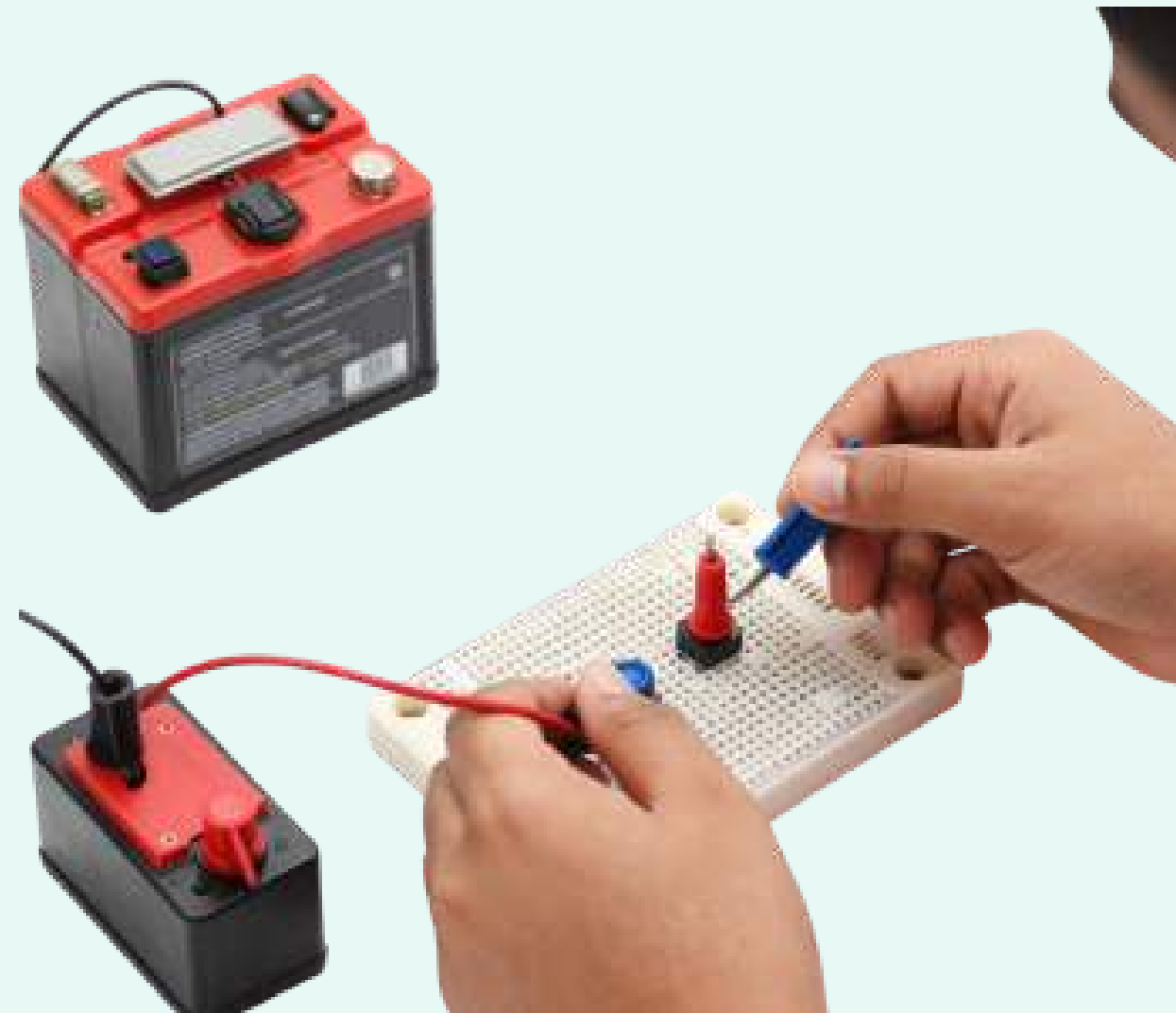


Battery (+) --> Wire --> Push-button -->
Wire --> Buzzer --> Wire --> Battery (-)
This is a SERIES circuit!

Let's Build It! Step 1

Connect the RED wire from the battery (+)
to one leg of the push-button switch.

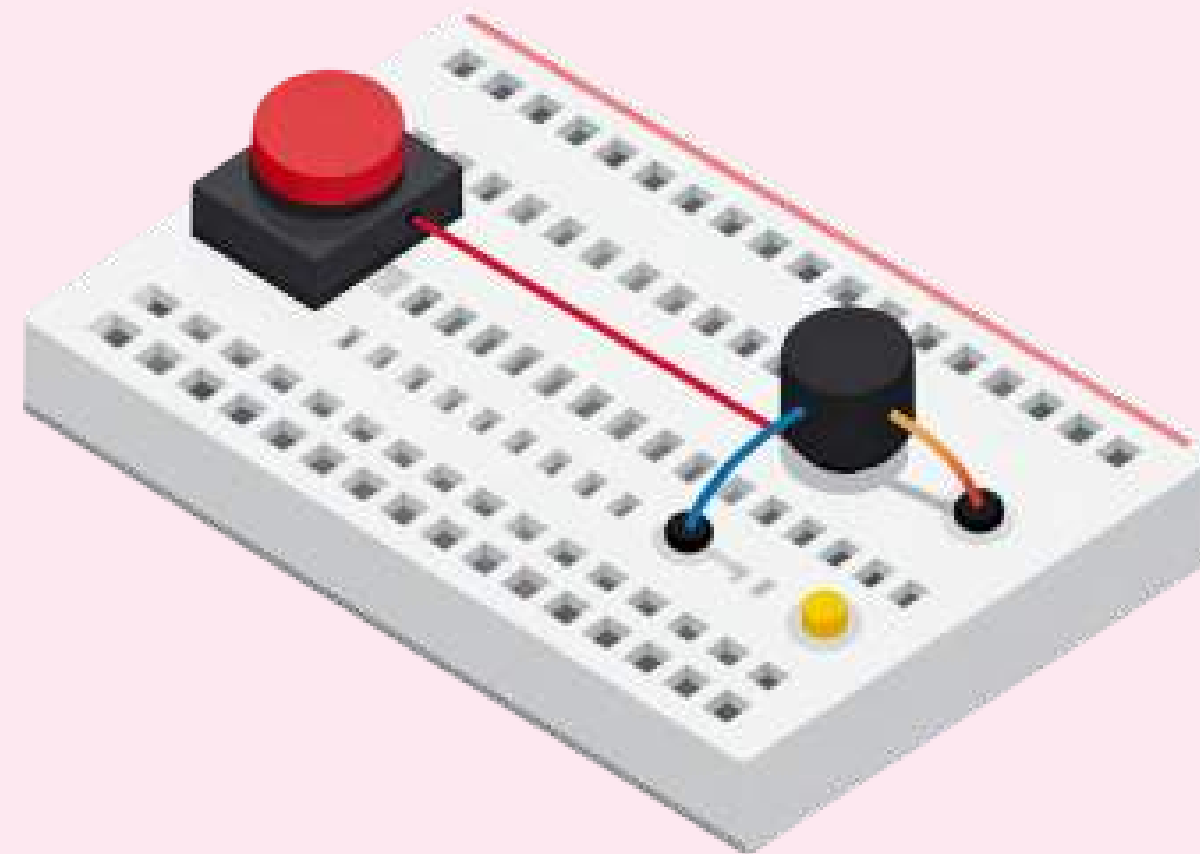
Red = positive!



Let's Build It! Step 2

Connect a wire from the OTHER leg of the push-button to the (+) pin of the buzzer.

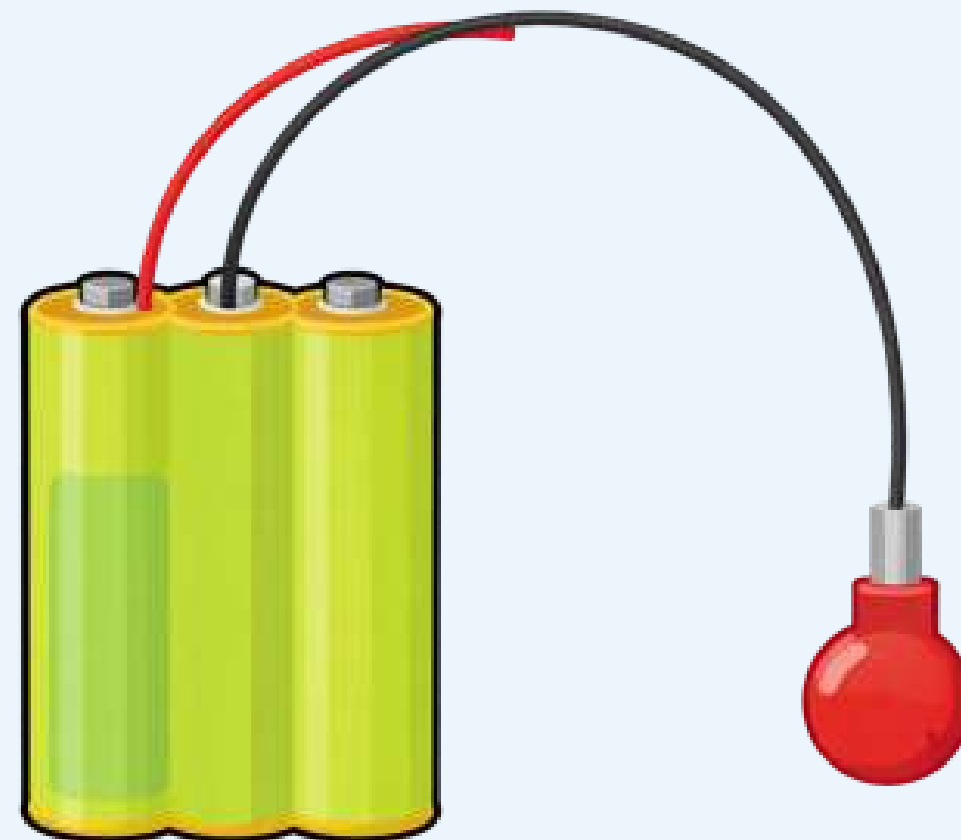
The switch is now in the middle!



Let's Build It! Step 3

Connect the BLACK wire from the buzzer (-)
back to the battery (-).

Circuit complete! Press the button!



What Happens?

Press the button --> buzzer goes BEEP!

Release the button --> silence.

You control the sound with your finger!



Real-life Examples



Doorbells use push-button switches!

Game show buzzers too!

Elevator buttons, keyboard keys...

Safety Rules

1. Use only low-voltage batteries (3V-9V)
2. Never connect to wall sockets!
3. Ask an adult if unsure
4. Handle wires carefully



Great Job! You Did It!

Today you learned:

- Push-button = momentary switch
- Buzzer = sound from electricity
- Series circuit connects them all!

Now try adding an LED too!

